

Artificial Intelligence in Real-World Systems

Introduction

Artificial Intelligence (AI) enables machines to simulate human intelligence such as learning, reasoning, and decision-making. AI is widely used in healthcare, banking, cybersecurity, e-commerce, and autonomous systems.

Real-Time Tech Challenges

1. Model hallucination in chatbots and LLMs. 2. High GPU cost for training deep learning models. 3. Bias in AI predictions due to poor datasets. 4. Latency issues in real-time AI applications. 5. Data privacy and compliance problems.

Real-Time Examples

- Netflix recommendation systems use machine learning to personalize content.
- Tesla uses AI for autonomous driving and object detection.
- Banks use AI fraud detection systems to monitor suspicious transactions.
- Hospitals use AI-assisted diagnostics for disease prediction.

Modern AI Stack

Python, TensorFlow, PyTorch, Hugging Face, LangChain, Vector Databases, and LLM APIs are commonly used in AI systems.

RAG Architecture

Retrieval-Augmented Generation (RAG) combines vector databases and LLMs to answer questions based on custom documents. Steps: 1. Load documents 2. Split into chunks 3. Create embeddings 4. Store in vector DB 5. Retrieve relevant chunks 6. Generate answer using LLM